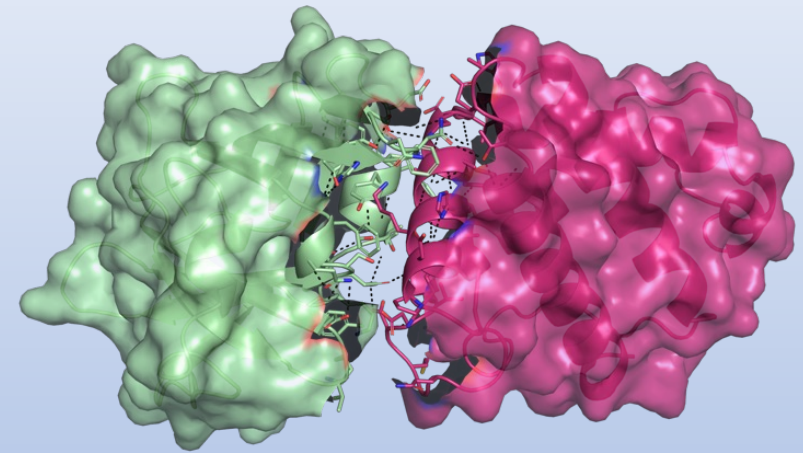
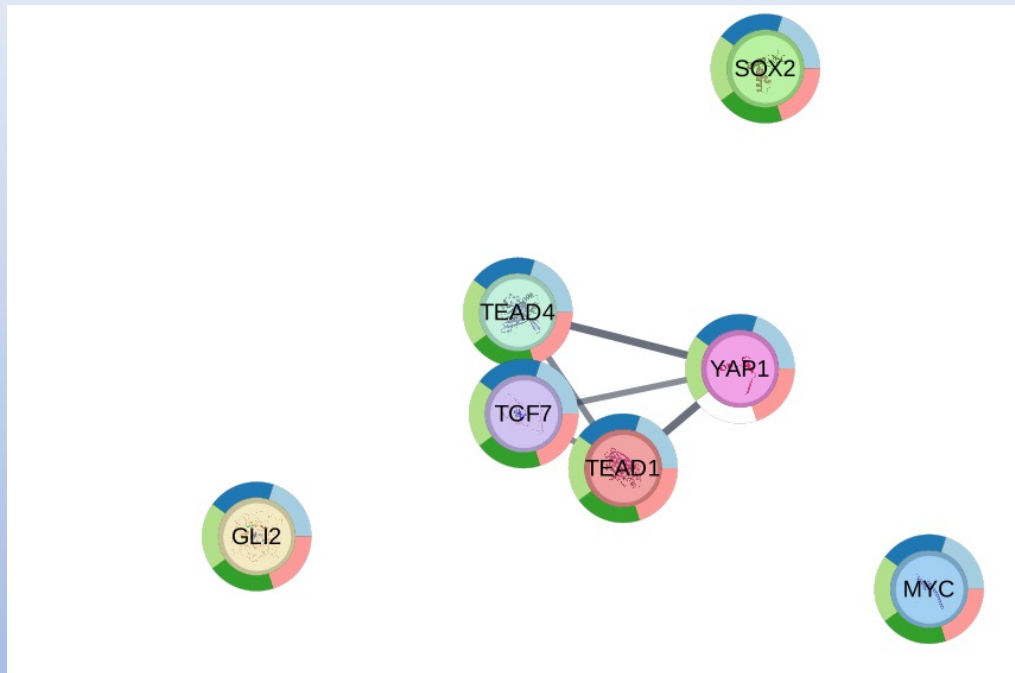


Modeling protein complex structures with AlphaFold 3



Prof Tiina A. Salminen



Structural Bioinformatics Laboratory

InFLAMES Research Flagship Center

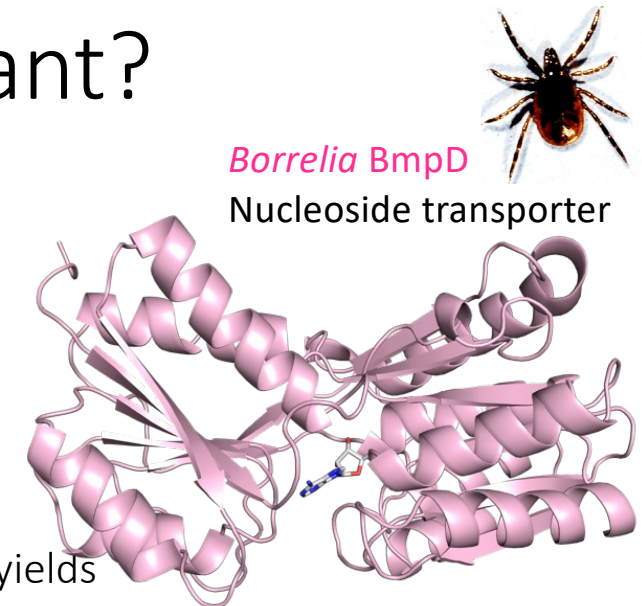
Biochemistry, Faculty of Science and Engineering

Why are protein structures important?

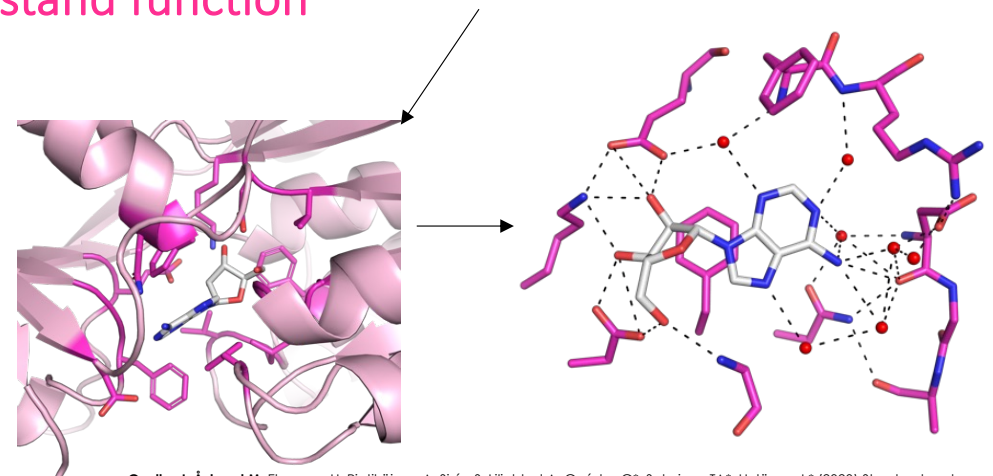
The function of a protein is determined by its structure!

- Structure provides insight into function
- Examples:
 - **Drug design:** therapeutic agents can be rationally designed
 - **Chemical industry:** more efficient enzymes can be engineered
 - **Agriculture:** modified proteins may increase stress tolerance and crop yields
- **In your project: analyze interactions to understand function**

Typically, ~10% of amino acid residues are functionally important, while ~90% maintain the 3D structure required for function



BmpD transports adenoside to *Borrelia*



Åstrand M, Cuellar J, Hytönen J, Salminen TA (2019) Predicting the ligand-binding properties of *Borrelia burgdorferi* s.s. Bmp proteins in light of the conserved features of related *Borrelia* proteins. *Journal of Theoretical Biology* 462:97-108.

Cuellar J, Åstrand M, Elovaara H, Pietikäinen A, Sirén S, Liljeblad A, Guédez G*, Salminen TA*, Hytönen J.* (2020) Structural and biomolecular analyses of *Borrelia burgdorferi* BmpD reveal a substrate-binding protein of an ABC-type nucleoside transporter family. *Infection and Immunity* 88(4):e00962-19.



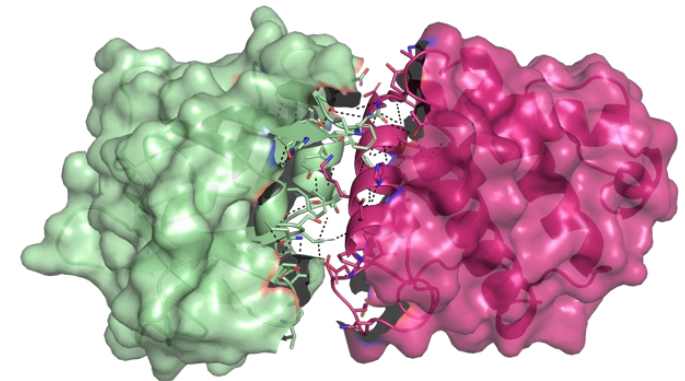
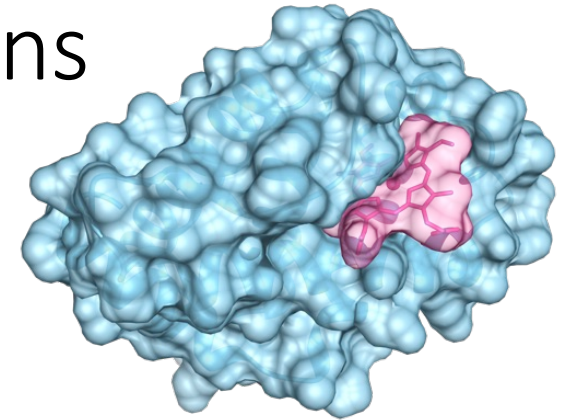
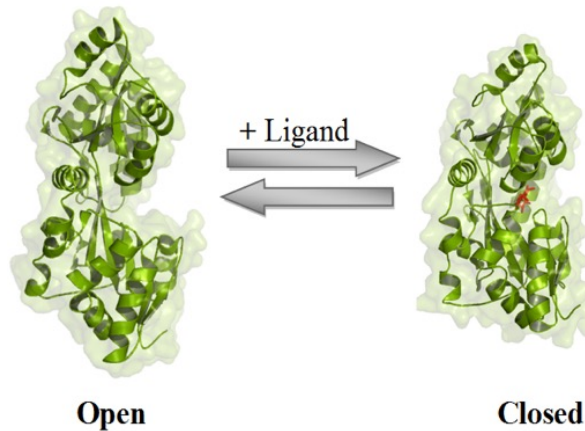
Instagram

#ääforskaren Mia Åstrand

Protein function depends on interactions with other molecules

Molecular interactions > Complex structures:

- Protein-Small molecule
- Protein-DNA
- Protein-Protein (PPI)



AlphaFold 3 modeling <https://alphafoldserver.com>

Protein-Protein Complexes:

Group 1 = TEAD1-TEAD4

Group 2 = YAP1 – TEAD4

Group 4 = YAP1 – TCF7

Group 5 = YAP1 – TEAD1

Protein-DNA Complexes:

Group 3 = MYC – DNA*

Group 6 = SOX2 – DNA (CTTTGTT)

*Study UNIPROT to find out the DNA sequence that binds your protein (Note: model the complex with DNA double helix!)

YAP-TEAD1

YAP-TCF7

YAP-TEAD4

TEAD1-TEAD4

MYC-DNA

SOX2-DNA

AlphaFold3 – Results

1. The 5 .cif files:

- **fold_<job_name>_model_<N>.cif**, where “<N>” is the rank of the predicted structure. Structures are ranked from 0 to 4, **where 0 has the highest confidence. – USE model 0**
- contain predicted structures in the mmCIF format.
- can be viewed in any molecular viewer like PyMOL or ChimeraX.

2. The 5 .json files:

- **fold_<job_name>_summary_confidences_<N>.json**, where “<N>” is the rank of the predicted structure from 0 to 4.
- contain summaries of the confidence metrics for the predictions
- **fold_<job_name>_full_data_<N>.json**, where “<N>” is the rank of the predicted structure from 0 to 4.
 - contain detailed confidence metrics, such as full PAE data, for the predictions

The **fold_<job_name>_job_request.json** file:

- contains the inputs of the modelling job and could be used to re-run the job

3. The terms_of_use.md file:

- a legal document detailing the terms of use for the predictions.

AlphaFold3 – Quality of prediction

pLDDT scores:

- a per-atom estimate confidence in a structure prediction
- 0-100 scale, where higher values indicate higher confidence:



pTM:

- Accuracy of the overall structure of the complex (each chain; complex)
- pTM > 0.5 the overall predicted fold for the complex might be true

ipTM:

- Accuracy of the predicted relative positions of the subunits forming the protein-protein complex
- ipTM > 0.8 confident high-quality prediction
- ipTM = 0.6 - 0.8 grey zone
- ipTM < 0.6 likely a failed prediction

AlphaFold3 – Quality of prediction

PAE (Predicted Aligned Error) scores:

Dark Green / Low PAE (< 5 Å):

High-confidence interaction

- The model is certain about how these two chains sit next to each other.

Light Green / Mid PAE (5 to 15 Å):

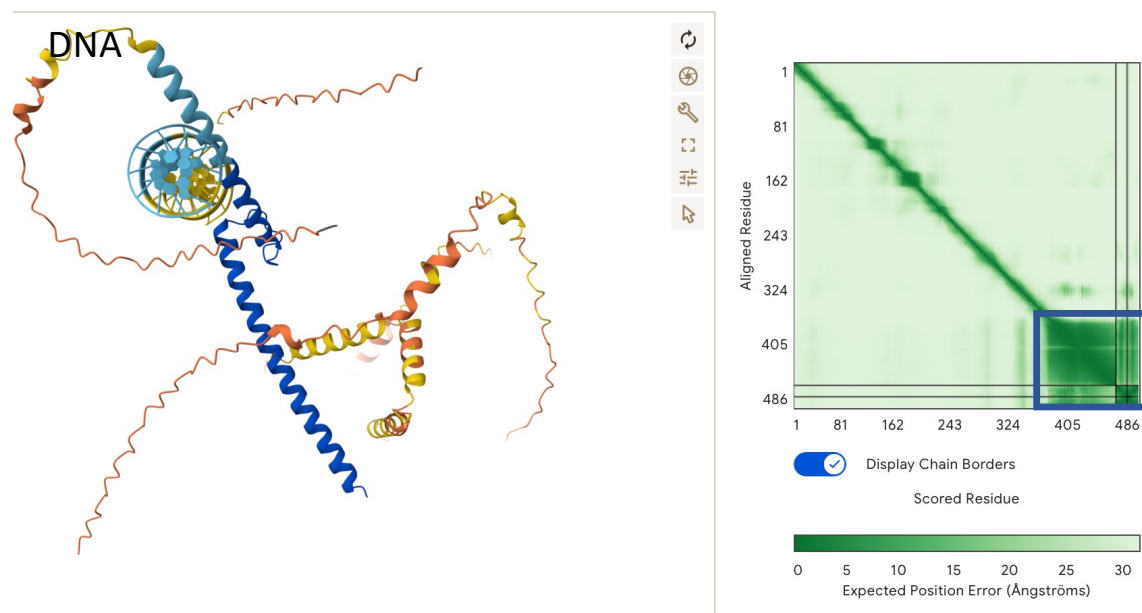
- The general orientation might be correct, but the exact surface contacts are blurry.

White or Yellow / High PAE (> 15 Å) :

Low confidence

- AlphaFold has no idea how two chains relate.
- They are floating randomly in the 3D model.

**NOTE: ALPHAFOLD QUALITY OF PREDICTION
= QUALITY OF *THE PROGRAM* PERFORMANCE**



Protein –DNA complex: The two C-terminal helices with high confidence (Blue) interact with the DNA with high-confidence (dark green PAE)